PRATEEK SAHU

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EDUCATION

University of Texas at Austin — *August 2017 - Present*
M.Sc/Ph.D., Computer Architecture. — GPA: 3.79/4
Indian Institute of Technology, Kanpur — *July 2011 - May 2015*
Bachelor of Engineering, Electrical. — GPA: 8.2/10

SELECT PUBLICATIONS

Sahu, P., Banerjee, S., Luo, M., Vahldiek-Oberwagner, A., Yadwadkar, N.J., Tiwari, M. (2024, Nov). *SoK: A systems perspective on compound ai threats and countermeasures* arXiv '24. [pdf]

Sahu, P., Wei, S., Yadwadkar, N.J., Tiwari, M. (2025, Mar). *Understanding Sidecars in Cloud Orchestration* SESAME '25, co-located with EuroSys '25. [pdf]

Behnia, M., **Sahu, P.**, Paccagnella, R., Yu, J., Zhao, Z., Zou, X., Unterluggauer, T., Torrellas, J., Rozas, C., Morrison, A., Mckeen, F., Liu, F., Gabor, R., Fletcher, C.W., Basak, A., Alameldeen, A. (2021, April). *Speculative Interference Attacks: Breaking Invisible Speculation Schemes* In Proceedings of the 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (pp. 1046-1060). [pdf]

Harris, A., Wei, S., **Sahu, P.**, Kumar, P., Austin, T., Tiwari, M. (2019, October). *Cyclone: Detecting Contention-Based Cache Information Leaks Through Cyclic Interference.* In Proceedings of the 52nd Annual IEEE/ACM International Symposium on Microarchitecture (pp. 57-72). ACM. [pdf]

WORK EXPERIENCE

Interests: Systems Security, Malware Detection, Secure AI
SPARK Reasearch Lab, University of Texas at Austin
Graduate Research Assistant, Advised by Prof. Mohit Tiwari — *Spring 2018 - Present*

- **Micro-architectural Side Channels**
  - *Speculative Interference Attacks* exploits younger instruction affecting older instruction latency.
  - Proof-of-concept attacks on caches that undermines defense mechanisms.
  - *Cyclone* is a side-channel malware detector that tracks cyclic contention events across security domains.
  - Evaluated against cache based side and covert channels on gem5 simulation environment.
- **Ransomware Detection**
  - *Lifecycle Aware Ransomware Detector* monitors cross layer system behavior across malware lifecycle.
  - Relies on hardware, system and network events to improve detection confidence.
  - Is trained on MITRE malware lifecycle stages for early detection and lower data loss.
- **Threat Modeling Generative AI systems**
  - Systematize cross layer threat vectors and model privileged adversaries in modern AI inference pipelines.
  - Explore system attack vectors to bypass application layer defense mechanisms to introduce prompt injection attacks.

Intel Labs, Intel Corp.
Graduate Technical Intern — *June 2020 - Nov 2020*

- **Secure Accelerator Design**: Design of secure data offload to Accelerators
  - System design for remote attestation
  - Proof of concept designs for key provisioning and secure data offload

SELECT COURSE PROJECTS

Verilog system design for 32-bit x86 ISA — *Microarchitecture Course, Dr. Y. Patt*
CPU design of a pipelined machine with memory & branch predictor for subset of x86 ISA. [Report]
Intelligent instruction duplication for side-channel defence — *Security Course, Dr. M. Tiwari*
Compiler solution for duplication of instructions which work on dummy data. [Report]
Low-power real-time object recognition SoC design — *SoC Design, Dr. A. Gerstlauer*
FPGA design and implementation of GEMM module of YOLO model. [Report]